Nathan Siu

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Education

University of California, Los Angeles — Los Angeles, CA
Ph.D. Candidate in Medical Informatics — GPA: 3.9 — *September 2021 – Expected December 2026*
Dissertation: Automating Biomarker Quantification in *Limbal Stem Cell Deficiency* Diagnosis and Treatment
Awards: Medical Informatics Fellowship ($41K); UCLA Bioscience Fellowship '22-'23 ($16K), '23-'24 ($19K)

Davidson College — Davidson, NC
B.S. in Bioinformatics — GPA 3.6 — *August 2017 – May 2021*
Thesis: A Tool to Quantify Localized Fluorescence in *Caenorhabditis elegans*
Awards: Bioinformatics Thesis Honors, Cum Laude, NIH R25 ($4K), 2020 ABRCMS Oral Presentation Award, 2020 UCLA Bruins-in-Genomics Research Excellence Award

Experience

UCLA Biomedical AI Research & Cornea Biology Labs — Los Angeles, CA
Graduate Student Researcher — *September 2021 – Present*

- Identified inefficiencies in a legacy diagnostic pipeline supporting a Phase I clinical trial, auditing existing workflows, ensuring data cleanliness, designing a technical solution, resulting in 5 conference presentations and 6 peer-reviewed publications
- Built and deployed AI-enabled tools (automation, computer vision/image processing pipelines, image segmentation/biomarker quantification) replacing time-consuming manual review steps, improving measurement reproducibility, and accelerating the trial timeline, saving hundreds of hours of labor
- Designed and facilitated multiple user studies to evaluate deployed software to identify UX pain points, assess inter-rater variability, and tailor tools to meet usability and measurement reproducibility specifications
- Led ongoing collaboration across a multi-disciplinary team, presenting findings to translate prototypes into deployed, validated tools; research contributed to multi-year funded NIH R01 grant ($600K)

United Auto Workers Local 4811 — Los Angeles, CA
Academic Student Employee Head Steward — *November 2022 – Present*

- Coordinated a statewide coalition-building and negotiation effort across the 10-campus University of California system, mobilizing hundreds of student-workers to secure a record $14.5M settlement
- Designed and executed a strategic advocacy campaign to secure formal institutional recognition of misclassified employees, winning additional contractual protections and benefits for over 300 workers
- Built a campus-wide engagement and onboarding program that scaled organizational membership to a supermajority, now representing over 60,000 workers, the largest of its kind in U.S. higher education

Davidson College Technology and Innovation — Davidson, NC
Student Technology Consultant — *August 2018 – May 2021*

- Managed end-to-end malware detection and remediation workflows across a fleet of college-provisioned devices, and led a proactive encryption campaign to standardize data security protocols for faculty and staff
- Authored and maintained a knowledge base of technical documentation, ensuring accuracy, accessibility, and consistency of information across a campus environment
- Assisted students, faculty, and staff in troubleshooting devices and cloud services, resolving hundreds of open tickets and cases

Davidson College Center for Teaching and Learning — Davidson, NC
Bioinformatics Programming Embedded Tutor — *January 2021 – May 2021*

- Designed and delivered structured practice materials for bi-weekly Python programming lab sessions, translating complex computational concepts into accessible, application-focused learning content
- Supported student learning outcomes through individualized instruction to convey technical material to audiences with varying levels of prior knowledge

Experience (continued)

Relay for Life / Colleges Against Cancer Davidson, NC
Treasurer *August 2017 – March 2020*

- Organized large scale fundraising events, managed cash flow, and coordinated activities with stakeholders on and off campus, collecting over $50K in donations for the American Cancer Society
- Successfully planned and executed multiple Relay for Life events with over 200 annual participants

University of Copenhagen Department of Public Health Copenhagen, Denmark
Research Assistant *August 2019 – December 2019*

- Studied the effectiveness of the Danish Capital Region's mammography screening program, resulting in co-authorship in a high-impact journal
- Awarded a travel grant for an academic visit to the Norwegian Cancer Registry and Norwegian Cancer Foundation in Oslo

Selected Works

Siu, Nathan, M. Vinet, S. Gonzalez, M. Ruiz, M. Canter, et al. (2026). "A Novel Pipeline for Adapting Cellpose for Segmentation and Quantification of Cultivated Limbal Stem Cells". In: *(Under review) Translational Vision Science & Technology*.

Siu, Nathan, M. Vinet, P. Shettiwar, T. Tran, K. Chen, et al. (2026). "Automated Image Registration Method for In Vivo Confocal Microscopy of the Corneal Sub-basal Nerve Plexus". In: *AMIA Annu Symp Proc* 2025, pp. 1209–1218.

Vinet, M., **Siu, Nathan**, C. Bonnet, C. Choi, S. Wu, et al. (2026). "Automated Quantification of Central Corneal Epithelial Thickness for the Diagnosis of Limbal Stem Cell Deficiency". In: *PLOS ONE* 21.10, e0358258.

Siu, Nathan, M. Vinet, C. Bonnet, W. Speier, and S. X. Deng (2025). "Automated Segmentation of Central Corneal Basal Cells from In Vivo Confocal Microscopy Imaging". In: *Invest. Ophthalmol. Vis. Sci.* 66.8, pp. 572–572.

Vinet, M., **N. Siu**, C. Bonnet, W. Speier, and S. X. Deng (2025). "Epithelial Layer Segmentation of Anterior Segment Optical Coherence Tomography in Eyes with Limbal Stem Cell Deficiency". In: *Invest. Ophthalmol. Vis. Sci.* 66.8, pp. 583–583.

Pett*, L., B. Hollenberg*, J. Mahoney*, J. Paz*, **N. Siu***, et al. (2023). "Participation and adherence to mammography screening in the Capital Region of Denmark: The importance of age over time". In: *PLOS ONE* 18.1, e0280790.

Liu*, P., S. Panchavati*, M. Pleasure*, **N. Siu***, C. Bonnet, et al. (2022). "MobileNetV2 Based Diagnosis and Grading of Limbal Stem Cell Deficiency". In: *2022 IEEE 22nd International Conference on Bioinformatics and Bioengineering (BIBE)*, pp. 174–179.

Siu, Nathan, M. Ruiz, S. G. Garrido, Y. Yan, D. Steinecke, et al. (2021). "Automatic Estimation of Limbal Stem Cell Densities in Cultured Epithelial Cell Microscopy Imaging". In: *2021 IEEE 21st International Conference on Bioinformatics and Bioengineering (BIBE)*, pp. 1–6.

More on Google Scholar.

Skills

Research Experience	multimodal medical AI, deep learning, computer vision, image processing, data science, data visualization, foundation models, agentic AI, public health, human subjects research, ophthalmology
Technical Skills	Python, PyTorch, scikit-learn, pandas, EDA, R, Unix shell, Java, d3.js, ImageJ Macro/Jython, git version control; Mandarin Chinese (native), French (B2)
Hobbies & Interests	pottery/ceramics, language learning, baking, reading, kombucha brewing